

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Mark | Comments                                                                                                     |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------|
| 07.1     | 1. Lowers activation energy;<br>2. Induced fit <b>causes</b> active site (of enzyme) to change shape;<br>3. (So) enzyme-substrate complex <b>causes</b> bonds to form/break;                                                                                                                                                                                                                                                                                                     | 3    | 3. Accept: description, of induced fit<br>3. Accept: enzyme-substrate complex causes stress/strain on bonds. |
| 07.2     | Size/dimensions /mass/variety of potato<br><b>OR</b><br>Temperature (of solution/flask)<br><b>OR</b><br>pH (of solution);                                                                                                                                                                                                                                                                                                                                                        | 1    | Accept : weight of potato<br>Ignore : amount of potato<br>Ignore concentration/ volume of catalase           |
| 07.3     | 0.33, 0.60, 0.86, 1.0, 1.0 = 2 marks;;<br>$\frac{6}{time}$<br>2 significant figures<br><br>If answer incorrect accept for 1 mark,<br>Correct values but incorrect number of significant figures<br><b>OR</b><br>1.0 written on row for hydrogen peroxide 2.0/2.5 in Table 5<br><b>OR</b><br>Answers showing correct division, eg 0.3, 0.6, 0.9<br><b>OR</b><br>Answers showing correct significant figures using incorrect calculation ( $\div 18$ ) 1.0, 0.56, 0.39, 0.33, 0.33 | 2    |                                                                                                              |

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| 07.4  | 1. Hydrogen peroxide concentration on x axis <b>and</b> rate of reaction on Y axis, linear number sequence <b>and</b> appropriate scale;<br><br>2. Correct units /mol dm <sup>-3</sup> <b>and</b> /arbitrary units/au;<br><br>3. All co-ordinates plotted accurately <b>with</b> point-to-point or smooth curve; | 3  | 1. Graph should cover half or more of the grid; eg reject if Y axis covers only three big squares<br><br>2. Accept brackets instead of solidus<br><br>3. Accept accurate plotting of co-ordinates given in 07.3<br><br>3. Reject: bar chart<br>3. Reject: if ruled straight line of best fit<br>3. Accept: if x axis starts at 0.5<br><br>3. Accept: if line is extended to (0,0)<br><br>Plot coordinates must be processed data, hydrogen peroxide vs time = 0 |
| 07.5  | Cut up/use discs/homogenise/increase surface area (of potato chips)<br><b>OR</b><br>Use bigger chips<br><b>OR</b><br>Increase temperature<br><b>OR</b><br>Change pH;                                                                                                                                             | 1  | Reject answer if the temperature is above 40°C<br><br>Ignore: more/increase heat                                                                                                                                                                                                                                                                                                                                                                                |
| TOTAL |                                                                                                                                                                                                                                                                                                                  | 10 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |